

Report:

Part 1: The Project Simplified

Imagine you are building the "Recommended for You" section of Netflix or Amazon.

Normally, if you have 1,000 users and 1,000 movies, that's 1 million possible combinations. A standard database struggles with that. In this project, you will use **Databricks** (a cloud platform) and **Apache Spark** (an engine for big data) to:

1. **Store** the data in the cloud (S3 or Data Lake).
2. **Clean** the data so it's ready for the computer to read.
3. **Run an ALS Model:** This is a "Collaborative Filtering" algorithm. It basically says: *"If User A and User B both liked 'Iron Man,' and User A also liked 'The Avengers,' then User B will probably like 'The Avengers' too."*

Part 2: Task Distribution (5 Students)

To finish in 3 weeks, we will treat this like a professional "Data Squad."

- **Student 1: Cloud Architect (The Foundation)**
 - Set up the Databricks workspace and link it to the cloud storage (AWS S3 or Azure).
 - Responsible for **Data Ingestion:** Getting the CSV files into a Spark DataFrame.
- **Student 2: Data Engineer (The Cleaner)**
 - Handles **Preprocessing:** Removing null values and normalizing data.

- Ensures the data types are correct (e.g., UserID must be an integer, not text).
 - **Student 3: Data Analyst (The Explorer)**
 - Performs **EDA**: Creates charts to show which movies are most popular and identifies "sparsity" (how many users haven't rated anything yet).
 - **Student 4: ML Engineer (The Builder)**
 - Implements the **ALS Model** using Spark MLlib.
 - Fine-tunes the "Rank" and "Iterations" of the model.
 - **Student 5: Specialist & Evaluator (The Finisher)**
 - Handles the "**Choose One**" task (e.g., K-Means Clustering to group similar users).
 - Calculates the **RMSE** (Error rate) to see how accurate the model is.
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Part 3: The 3-Week Timeline

Week 1: Setup & Data Pipeline

- **Goal:** Get the data living in the Cloud.
- **Deliverable:** A Databricks notebook where you can successfully "Select *" from your dataset.
- **Key Task:** Connect your Cloud Storage to Databricks so you aren't uploading files manually from your PC every time.

Week 2: EDA & The Baseline Model

- **Goal:** Understand your data and build the first version of the recommender.

- **Deliverable:** A working ALS model that can predict a rating for a specific user.
- **Key Task:** The Data Analyst and ML Engineer must work together here to ensure the model isn't trained on "garbage" data.

Week 3: Advanced Features & Results

- **Goal:** Add the "Specialty" (Clustering) and generate the final list.
- **Deliverable:** A Top-N Recommendation list (e.g., "Top 5 Movies for User #402").
- **Key Task:** Calculate the **RMSE**. If the error is high, adjust the model settings one last time

Project File Structure

I recommend creating a main folder in Databricks called Safe_Vision_RecSys with the following structure:

-  **01_Ingestion/**
 -  01_Cloud_Setup.ipynb — **(Student 1):** Connects S3/Data Lake and loads raw CSVs.
-  **02_Processing/**
 -  02_Data_Cleaning.ipynb — **(Student 2):** Cleans the data and saves it as a "Silver Table."
 -  03_Exploratory_Analysis.ipynb — **(Student 3):** Creates charts and stats from the Silver Table.

- 📁 **03_Modeling/**
 - 📄 04_ALS_Training.ipynb — **(Student 4)**: Trains the recommendation engine.
 - 📄 05_User_Segmentation.ipynb — **(Student 5)**: Clusters users and calculates final accuracy (RMSE).
- 📁 **04_Production/**
 - 📄 06_Final_Dashboard.ipynb — **(All)**: Displays the Top-N recommendations for a user.

🚧 The "Blocker" Map (Dependencies)

In Cloud Database projects, we use a **Pipeline**. If the person before you doesn't finish, you have no data to work with.

1. The Ultimate Blocker: Student 1

- **Who they block:** Everyone.
- **Why:** If Student 1 doesn't successfully connect the Cloud Storage and load the CSV into Databricks, nobody else has a dataset to see.
- **Solution:** Student 1 should aim to finish by the middle of Week 1.

2. The Quality Blocker: Student 2

- **Who they block:** Student 3, Student 4, and Student 5.
- **Why:** Student 2 "cleans" the data. If Student 4 tries to train a model on "dirty" data (missing values or wrong formats), the code will crash.
- **Dependency:** Student 2 must wait for Student 1 to finish the ingestion.

3. The Parallel Path: Student 3

- **Who they block:** Nobody directly, but they support the ML team.
- **Why:** Student 3 can work at the same time as Student 4. However, they need Student 2 to finish the cleaning first so their charts are accurate.

4. The Logic Blocker: Student 4

- **Who they block:** Student 5.
- **Why:** Student 5 is doing the "Bonus" evaluation and user clustering. They need the "Predictions" generated by Student 4's model to calculate the RMSE (error rate).